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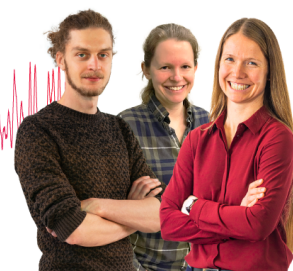
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Calculation of the Quadruple Deformation Parameter (β_2) and Reduced Transition Probability $B(E2; \uparrow)$ of some Even-Even Semi-magic Nuclei with Neutron Numbers $N=28, 50$ & 82

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Abstract. In this work, the quadruple deformation parameter (β_2) and the reduced transition probability $B(E2; \uparrow)$ for the transition ($0_1^+ \rightarrow 2_1^+$) have been calculated for the semi-magic nuclei with magic neutron numbers, which are (^{44}S , ^{46}Ar , ^{48}Ca , ^{50}Ti , ^{52}Cr , ^{54}Fe) for $N=28$, (^{84}Se , ^{86}Kr , ^{88}Sr , ^{90}Zr , ^{92}Mo , ^{94}Ru , ^{96}Pd , ^{98}Cd , ^{56}Ni) for $N=50$ and (^{132}Sn , ^{134}Te , ^{136}Xe , ^{138}Ba , ^{140}Ce , ^{142}Nd , ^{144}Sm , ^{146}Gd , ^{148}Dy , ^{150}Er , ^{152}Yb , and ^{154}Hf) for $N=82$ have been calculated using two different methods. The first method we used the global best fit systematic method has been utilized to determine the values of (β_2) and the second one is the nucleus deformation parameter δ from the quadrupole moment Q_0 . The Quadruple deformation parameter (β_2) for the studied nuclei using the mentioned methods were plotted as a function of the atomic number Z . The behavior of the calculated deformation parameter (β_2) was discussed and shows that it decreasing till minimum values at $Z=20$ then it tends to increasing for the semi-magic nuclei with neutron magic number $N=28$. The deformation parameter (β_2) for nuclei with $N=50$ neutron magic number shows same behavior of nuclei with $N=28$ and the minimum were found at $Z=40$ then increase and decrease again. The semi-magic nuclei at $N=82$ shows increase and then gradual decrease. of the semi-magic nuclei with $N=28$ decreases with the increase in the semi-magic number of neutrons. The lowest deformation values were at $N=82$, while the highest was at $N=28$.

Keywords Terms— Nuclear deformation, Semi-magic numbers, Global best fit, Quadruple Deformation parameter.

INTRODUCTION

The shape of the nuclei is one of the basic properties of it. Nuclei have a closed shell when they contain a magic number of protons (Z) and neutrons (N). Whereas, they have deformed shape when they have numbers of (Z) / (N) far from the magic numbers [1]. Figure 1 shows a schematic drawing of a deformed nucleus. The shape is determined by the relationship between the semi-axes a , b and c [2].

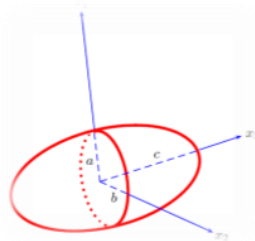


FIGURE 1. Depicts a deformed nucleus with intrinsic frame ($x_1 x_2 x_3$) and the semi-axes (a, b and c)

When ($a \neq b \neq c$) the deformed nucleus takes an ellipsoidal shape, this is corresponding to the “axially-symmetric triaxial” shapes that shown in Fig. 1. [2]. On the other hand, the case ($a = b \neq c$) corresponds to an “axially-symmetric” shape as shown in figure 2.2(c). We can define the deformation parameter (δ), in the case of axial-symmetry according to the equation below [2].

$$\delta = 2 \frac{c-b}{c+b} \quad (1)$$

See Fig. 2.2(a,b), prolate and oblate shape corresponds to ($\delta > 0$, $\delta < 0$) respectively

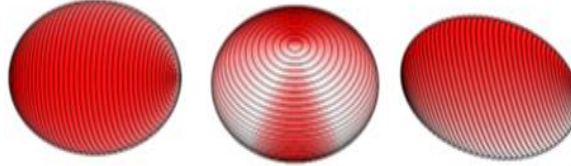


FIGURE 2. Illustrates the shapes of the deformed nucleus.

In general, the nuclear deformation take place from the method that valence nucleons organization themselves in an unfilled shell (i.e the deformation originates only when both neutron and proton shells are somewhat full)[3]. The simplest types of deformation are termed the quadrupole deformations. In such type of deformation, the nucleus can take an oblate and/ or prolate shape [1]. S. Akkoyunet *et al.* in (2015) estimated the values of reduced transition probability $B(E2) \uparrow$ for some even-even nuclei at ($110 \leq A \leq 190$) via utilizing “artificial neural network technique” [1]. It has been observed that the prediction power of this technique for Nd, Gd, Ce, Dy, Sm and Ba nuclei were high. In other words, ANN technique is found to be beneficial in order to evaluate $B(E2) \uparrow$ values of even-even nuclei in this region. Moreover, the nuclear magic numbers show maximum deviations[1]. Ali A. Alzubadi and Ali A. Abdulhasan in (2015) has been investigated the behavior the potential energy curves as a function of the quadrupole deformation parameter (β_2) for Sm, Na, Si and Cr isotopes within the framework of Skyrme-Hartree-Fock-Bogolyubov (SHFB) way. Using HFBTHO code which depend on the (SHFB) way, the calculation was performed[4]. Naz .T. Jarallah and Hussain J. Hassan in (2016) determined the shape for even-even ($_{54}\text{Xe}$ $118 \leq A \leq 140$ and $_{82}\text{Pb}$ $204 \leq A \leq 210$) nuclei from the deformation parameters δ and β_2 . These parameters were calculated utilizing two ways including: nucleus quadrupole deformation parameters δ from quadrupole moment (Qo) and nucleus quadrupole deformation parameter (β_2) from reduced transition probability $B(E2; \uparrow)$ for $0^+ \rightarrow 2^+_{\uparrow}$ transitions [5]. Huda A. Marid et al. in (2018) evaluated the eccentricity (e) and the deformation parameter (β_2) in order to study the relationship of the deformation of the nucleus to the radioactivity of ^{232}Th and ^{238}U chain. The results appeared that the radioactive nuclei decay into (α & β) in order to obtain a more stable state via changing their shape towards the spherical shape [6].

SEMI-MAGIC NUMBERS

The main idea behind the existence of semi-magic nuclei numbers is that a novel gap appears in the "single-nucleon level" diagram via adding a pair of protons or neutrons and its vanishing via adding another pair. This effect is believed to be due to the reaction among valence protons and neutrons. In the shell model, A. De Shalit and M. Goldhaber surmised the significance of the (proton- neutron) reaction in connection with the appearance of nuclear deformation. On the other hand, S. Pittel and P. Federman showed that the reaction between protons and neutrons with powerfully overlapping orbitals probably causes the appearance of nuclear deformation. Simultaneously, they showed that this reaction may fundamentally alter the "single-nucleon level" diagram. When the reaction of a neutron and a proton with large or radial quantum numbers is close to the quantum orbital, this interaction becomes important. So, semi-magic quantum numbers are fewer stable than classical numbers [7].

The aim of this study, is to determine the nucleus quadrupole deformation parameter (β_2) from reduced transition probability $B(E2; \uparrow)$ for the some even-even semi-magic nuclei with magic number of neutrons ($N=28, 50$ & 82) by utilizing the global best fit systematic approach.

THEORETICAL BACKGROUND

A. Reduced Transition Probability B(E2) For Gamma-Rays

The reduced transition probability $B(E2; \uparrow)$ among ground state 0^+ and first excited state 2^+ in even-even nuclides is one of important quantities since it involves a fundamental nuclear information around the energy of low lying levels in nuclei. The values of the $B(E2; \uparrow)$ confirms the occurrence of quadruple deformations of the nuclei (i.e related to nuclear quadruple deformation parameter (β_2)) [1].

Generally, it is measured via Coulomb excitation, Mössbauer spectroscopy, inelastic electron scattering, muonic x-ray measurement, resonance fluorescence or lifetime measurement [8]. Moreover, there are numerous theoretical model to calculation of the reduced transition probability $B(E2; \uparrow)$ values including relativistic mean-field model, finite-range droplet model, dynamical microscopic model, Woods-Saxon model and single-shell asymptotic Nilsson model etc. [9–13]. In present work, the values of $B(E2)$ are predicted according to global best Fit systematic from the following equation

$$B(E2) \uparrow = \frac{2.6 Z^2}{E_{\gamma_0} A^{2/3}} \quad (2)$$

Here

Z: Atomic number

E_{γ_0} : Gamma energy transition

A: Nuclide mass number

B. Quadruple Deformation parameter β_2

The intrinsic quadrupole moment of an evenly charged ellipsoid can be described, as is well known (for instance, see [14]).

$$Q_0 = 2/5 Z (b^2 - a^2) \quad (3)$$

where the small and large ellipsoid axes, respectively, are a and b. Since the degree of nucleus shape deviation from a sphere, the quadrupole deformation parameter δ , can be expressed as

$$\delta = 0.3 (b^2 - a^2) / 2 \langle r^2 \rangle \quad (4)$$

where

$$\langle r^2 \rangle = (b^2 + 2a^2)/5. \quad (5)$$

This is how Eq.(4) would be expressed.

$$Q_0 = 3/4 Z \langle r^2 \rangle \delta \quad (6)$$

Utilizing experimental quadrupole moment Q_0 values, therefore the formula could be used to determine the values of the nucleus quadrupole deformation parameter.

$$\delta = 0.75 Q_0 / (Z \langle r^2 \rangle) \quad (7)$$

The static electric quadrupole moment Q values observed in the laboratory system were recalculated [14] to yield intrinsic moment Q_0 values. Since it was assumed that the amplitude of near-spherical nucleus shape vibrations was less than that of nucleus equilibrium deformation, the following equation can be used

$$Q = Q_0 (3 K^2 - I(I + 1)) / ((I - 1)(2I + 3)), \quad (8)$$

where I is the state's spin and K is its projection on the symmetry axis. I: is a member of the rotational band based on the nucleus ground state.

The following expressions were used to calculate the value of $\langle r^2 \rangle$.

$$\langle r^2 \rangle = 0.6 R_0^2 (1 + 10/3 (\pi a_0 / R_0)^2 / (1 + (\pi a_0 / R_0)^2)) (A \leq 100), \quad (9)$$

$$\langle r^2 \rangle = 0.6 (1.2 A^{1/3})^2 (A > 100), \quad (10)$$

which considers the effects of the surface diffusion characteristics of light nuclei.

The information on fast electron scattering was used to calculate the parameters of the radial Woods-Saxon potential form-factor ($R_0 = 1.07 A^{1/3}$ fm and $a_0 = 0.55$ fm).

It is obvious that the deformation parameter (β_2) is described the shape of the axially symmetric deformed nucleus (i.e. measures elongation of the axially symmetric shape) [14,15]. This parameter is directly connected to the reduced transition probability $B(E2)$ according to the relationship below [2,16]:

$$\beta_2 = \frac{4\pi}{3ZR_0^2} \left[\frac{B(E2;\uparrow)}{e^2} \right]^{1/2} \quad (11)$$

Here

R_0^2 : The average nuclear radius, which is calculated from the relationship below:

$$R_0^2 = 0.0144 A^{2/3} \text{ barn} \quad (12)$$

Furthermore, the deformation parameter (β_2) is highly related to quadrupole moment (Q_0) (i.e. homogeneous charge distribution) by the formula [14,15]

$$\beta_2 = \frac{Q_0 \sqrt{5\pi}}{3ZR_0^2} \quad (13)$$

RESULTS AND DISCUSSIONS

In this work, the global best fit systematic have been utilized in order to evaluate the reduced transition probability ($B(E2;\uparrow)$) rely on the equation (2) in units of (e^2b^2) and is presented as (theoretical work). Furthermore, we have determined the quadruple deformation parameter (β_2) of some even-even semi magic nuclei with ($40 < A < 160$) according to the equation (12) and referred as (present work). The results of the calculated deformation parameter (β_2) are tabulated in Tables (1-3). In order to study the relationship between deformation parameter (β_2) and semi-magic nuclei with neutron magic number at $N=28, 50$, and 82 . The quadruple deformation parameter (β_2) were plotted using graphed software as a function of the atomic number (Z) for the studied nuclei are depicted in figures (3-5). On the other hand, nuclei with neutron number (N) far from magic number are generally deformed. These figures demonstrate that the deformation parameters decreased as they got closer to the neutrons magic number. It means that nuclei that contain special numbers of neutrons have a "closed shell" that promotes a spherical shape. Additionally, we can see plainly from these figures that the values of β_2 are larger than the values of for all of our findings, indicating that the deformation resulting from the transition probability $B(E2;\uparrow)$ is greater than the deformation resulting from Q_0 .

The difference between calculated deformation parameter β_2 from the global fit method of $B(E2;\uparrow)$ is due to the hypothesis that values of do not account for such nucleus surface vibrations, whereas values of β_2 calculated from the quadruple moment depends on the surface vibration. In other words, the values of β_2 are affected by both dynamic and static nucleus deformation, which resulted from the nucleus' surface vibration.

The theoretical values for $B(E2;\uparrow)$ and β_2 [8] and our results (present work) are shown in Tables 1, 2 and 3, for $N=28, 50$ and 28 , respectively. For $B(E2;\uparrow)$ and β_2 , these values clearly differ from each other, and this difference is due to the fact that we used the Global Best Fit equation to calculate $B(E2)$, which is used to calculate the deformation parameter 2, while the reference data uses the adopted values [8] for $B(E2;\uparrow)$. On the other hand, it is seen that the quadruple deformation parameter (β_2) decreases with increase in number of neutrons at the expense of the number of protons (i.e. the increase in the number of protons increases the Coulomb repulsion force and this in turn affects the shape of the nucleus and makes it more deformed).

TABLE 1: Nuclide mass number, atomic number Z , gamma energy transition E_{γ_0} , reduced transition probabilities $B(E2) \uparrow e^2b^2$, deformation parameter β_2 and average nuclear radii R_0^2 for isotones nuclide with ($N = 28$).

Nuclei	Z	E_{γ_0} (keV) [14]	Theoretical values [8]		Present work		
			$B(E2) \uparrow e^2b^2$ Global Best Fit	β_2	$B(E2) \uparrow e^2b^2$	R_0^2 (b)	β_2
^{44}S	16	1315	0.040	0.25438	0.0406	0.1795	0.2940
^{46}Ar	18	1550	0.042	0.17518	0.0423	0.1849	0.2590
^{48}Ca	20	3831	0.0203	0.10618	0.0206	0.1902	0.1679
^{50}Ti	22	1553	0.059	0.16611	0.0597	0.1954	0.2380
^{52}Cr	24	1434	0.074	0.2235	0.0750	0.2006	0.2382
^{54}Fe	26	1408	0.086	0.1958	0.0874	0.2057	0.2315
^{56}Ni	28	2700	0.051	0.17317	0.0516	0.2108	0.1612

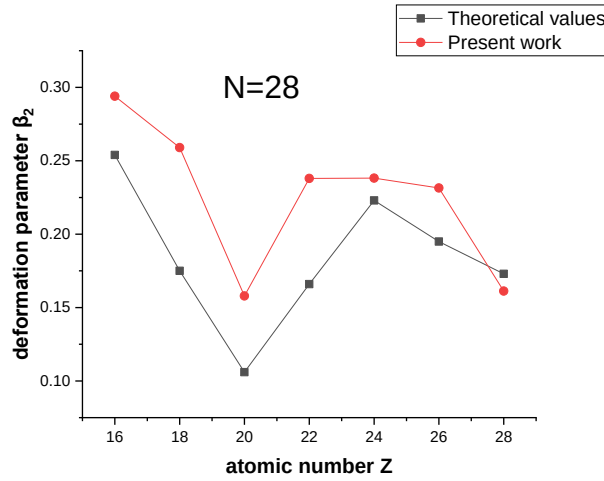


FIGURE 3. Deformation parameter β_2 as a function of atomic number for $N=28$ neutron magic number.

TABLE 2: Nuclide mass number, atomic number Z , gamma energy transition E_{γ_0} , reduced transition probabilities $B(E2) \uparrow e^2b^2$, deformation parameter β_2 and average nuclear radii R_0^2 for isotones nuclide with ($N = 50$).

Nuclide	Z	E_{γ_0} (keV) [14]	Theoretical values [8]		Present work		
			$B(E2) \uparrow e^2b^2$ Global Best Fit	β_2	$B(E2) \uparrow e^2b^2$	R_0^2 (b)	β_2
^{84}Se	34	1454	0.106	-	0.107	0.2762	0.1464
^{86}Kr	36	1564	0.109	0.145	0.1106	0.2806	0.1379
^{88}Sr	38	1836	0.102	0.1173	0.1034	0.2849	0.1244
^{90}Zr	40	2186	0.093	0.0894	0.0948	0.2892	0.1115
^{92}Mo	42	1509	0.147	0.1058	0.1491	0.2935	0.1312
^{94}Ru	44	1430	0.168	-	0.1703	0.2977	0.1320
^{96}Pd	46	1415	0.183	-	0.1854	0.3019	0.1299
^{98}Cd	48	1394	0.199	-	0.2022	0.3061	0.1282

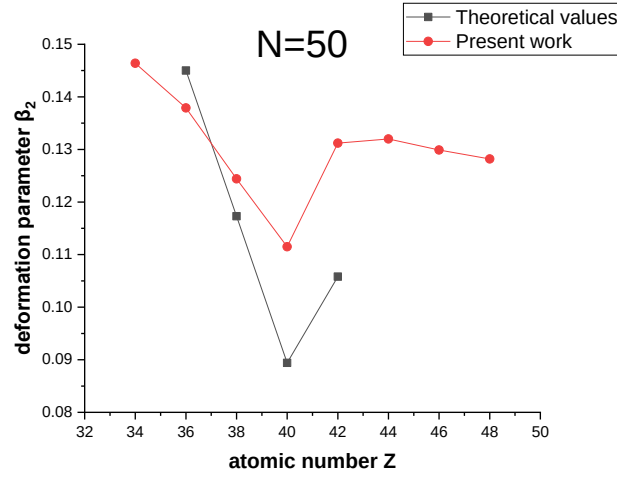


FIGURE 4. Deformation parameter β_2 as a function of atomic number for N=50 neutron magic number.

TABLE 3: Nuclide mass number, atomic number Z, gamma energy transition E_{γ_0} , reduced transition probabilities $B(E2) \uparrow e^2b^2$, deformation parameter β_2 and average nuclear radii R_0^2 for isotones nuclide with (N = 82).

Nuclide	Z	E_{γ_0} (keV) [14]	Theoretical values [8]		Present work		
			$B(E2) \uparrow e^2b^2$ Global Best Fit	β_2	$B(E2) \uparrow e^2b^2$	R_0^2 (b)	β_2
^{132}Sn	50	4041	0.061	-	0.0620	0.3733	0.0559
^{134}Te	52	1279	0.207	-	0.2099	0.3771	0.0979
^{136}Xe	54	1313	0.215	0.122 10	0.2183	0.3808	0.0952
^{138}Ba	56	1435	0.210	0.0933 18	0.2128	0.3845	0.0897
^{140}Ce	58	1596	0.201	0.1015 10	0.2033	0.3883	0.0839
^{142}Nd	60	1575	0.215	0.0917 10	0.2183	0.3919	0.0832
^{144}Sm	62	1660	0.216	0.0874 10	0.2191	0.3956	0.0799
^{146}Gd	64	1971	0.192	-	0.1949	0.3993	0.0724
^{148}Dy	66	1677	0.238	-	0.2414	0.4029	0.0774
^{150}Er	68	1578	0.266	-	0.2699	0.4065	0.0787
^{152}Yb	70	1531	0.29	-	0.2922	0.4101	0.0789
^{154}Hf	72	1513	0.31	-	0.3101	0.4137	0.0783

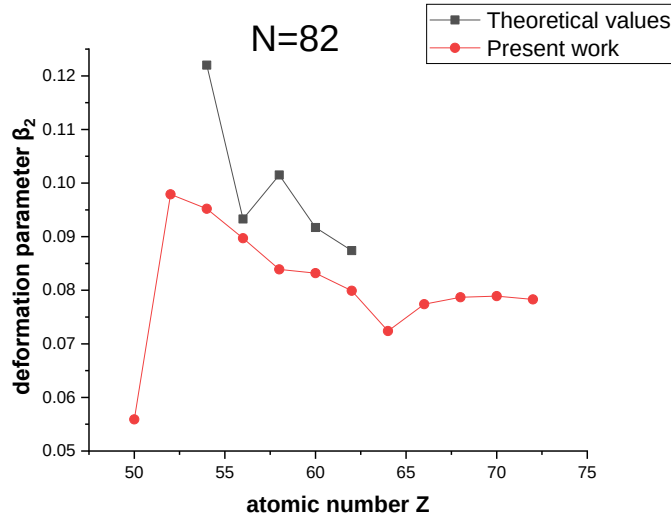


FIGURE 5.: Deformation parameter β_2 as a function of atomic number for N=82 neutron magic number.

CONCLUSION

The inputs values of reduced transition probability $B(E2; \uparrow)$ have been produced and assembled from various experimental data. The present study shows that the parameter β_2 decreases when the atomic number be closer to the protons semi magic number (i.e nuclei with semi magic numbers of protons have a "closed shell" that encourages a spherical shape). Furthermore, we conclude that the Quadruple Deformation Parameter (β_2) of the nuclei decreases with the increase in the semi-magic number of neutrons (the lowest deformation values were at N=82).

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